Chapter 7. Principal Component Analysis

Population and sample principal components, SVD, choosing the number of components

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Ch. 1, §6, introduced PCA through sample geometry. Here it is formulated at the population level, with the Lagrange-multiplier derivation, the SVD and the choice of the number of components.

1 Population Principal Components

Motivation. With many correlated variables, most of the variation often lies in a few directions. Principal components are the orthogonal directions of maximal variance, the eigenvectors of $\boldsymbol{\Sigma}$, and they allow us to summarize the data in fewer dimensions.

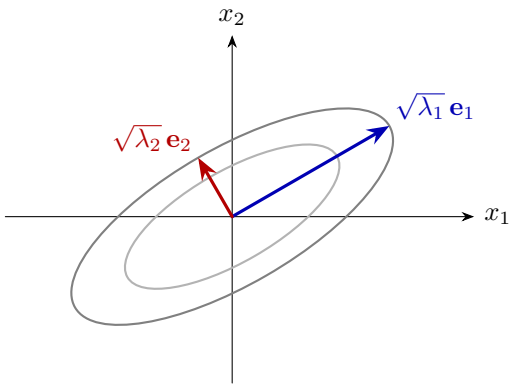


Figure. *Contours of a bivariate normal density are ellipses whose axes are the eigenvectors $\mathbf{e}_1, \mathbf{e}_2$ of $\boldsymbol{\Sigma}$, with half-lengths proportional to $\sqrt{\lambda_1}, \sqrt{\lambda_2}$. The first principal component is the projection on the long axis, the direction of maximal variance.*

Let $\mathbf{X} \in \mathbb{R}^p$ with $\mathrm{Cov}(\mathbf{X}) = \boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$, $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$, $\boldsymbol{\Lambda} = \mathrm{diag}(\lambda_1, \dots, \lambda_p)$, $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$. A linear combination $Z(\mathbf{a}) = \mathbf{a}^\top \mathbf{X}$ has variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$. (For $d = 1$: $\mathbf{a}_1 = \arg\max_{\|\mathbf{a}\|=1} \mathrm{Var}(Z(\mathbf{a}))$.)

Definition 7.1: principal components

First PC direction vector: $\mathbf{u}_1 = \arg\max_{\|\mathbf{u}\|=1} \mathrm{Var}(Z(\mathbf{u}))$, with $\lambda_1 = \mathrm{Var}(Z_1)$. Second: $\mathbf{u}_2 = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_1 = 0} \mathrm{Var}(\mathbf{u}^\top \mathbf{X})$, with $\lambda_2 = \mathrm{Var}(Z_2)$; and so on:

$$\mathbf{u}_k = \arg \max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top \mathbf{u}_j = 0\ (j<k)} \mathrm{Var}(\mathbf{u}^\top \mathbf{X}), \qquad Z_k = \mathbf{u}_k^\top \mathbf{X}.$$

Then $\lambda_1 \geq \lambda_2 \geq \cdots$ and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Theorem 7.2: solution

$\mathbf{u}_k$ is the kth eigenvector of $\boldsymbol{\Sigma}$, $\mathrm{Var}(Z_k) = \lambda_k$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$.

Lagrange multipliers. **1.** $\Phi(\mathbf{u}, \lambda) = \mathrm{Var}(\mathbf{u}^\top \mathbf{X}) - \lambda(\mathbf{u}^\top \mathbf{u} - 1) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1)$. $\frac{1}{2} \frac{\partial}{\partial \mathbf{u}} \Phi(\mathbf{u}, \lambda) = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} = \mathbf{0}$, so $\boldsymbol{\Sigma}\mathbf{u} = \lambda \mathbf{u}$ and $\lambda = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u}$: the objective equals the eigenvalue, so take the largest, λ_1.

2. With $\boldsymbol{\gamma}^k = (\gamma_1, \dots, \gamma_{k-1})$,

$$\Phi(\mathbf{u}, \lambda, \boldsymbol{\gamma}^k) = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} - \lambda(\mathbf{u}^\top \mathbf{u} - 1) - \sum_{j=1}^{k-1} 2\gamma_j \mathbf{u}_j^\top \mathbf{u},$$

$$\frac{1}{2} \frac{\partial \Phi}{\partial \mathbf{u}} = \boldsymbol{\Sigma}\mathbf{u} - \lambda \mathbf{u} - \sum_{j=1}^{k-1} \gamma_j \mathbf{u}_j = \mathbf{0}, \quad \frac{\partial \Phi}{\partial \gamma_j} = \mathbf{u}_j^\top \mathbf{u} = 0.$$

Premultiplying by $\mathbf{u}_i^\top$ $(i<k)$: $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u}=\gamma_i$, and $\mathbf{u}_i^\top\boldsymbol{\Sigma}\mathbf{u}=\lambda_i\mathbf{u}_i^\top\mathbf{u}=0$, so $\gamma_i=0$. Hence $\boldsymbol{\Sigma}\mathbf{u}=\lambda\mathbf{u}$ again, and among eigenvectors orthogonal to $\mathbf{u}_1,\dots,\mathbf{u}_{k-1}$ the largest eigenvalue λ_k is chosen: $\mathbf{u}_k,\lambda_k$. Finally $\mathrm{Cov}(Z_j,Z_k)=\mathbf{u}_j^\top\boldsymbol{\Sigma}\mathbf{u}_k=\lambda_k\mathbf{u}_j^\top\mathbf{u}_k=0$. ■

Theorem 7.3: total variance and correlations

1. $\sum_{j=1}^p \mathrm{Var}(X_j)=\mathrm{tr}\boldsymbol{\Sigma}=\sum_{k=1}^p\lambda_k=\sum_k\mathrm{Var}(Z_k)$; the proportion of variance explained by PC k is $\lambda_k/\sum_j\lambda_j$.
2. $\mathrm{Corr}(Z_k,X_j)=\dfrac{u_{jk}\sqrt{\lambda_k}}{\sqrt{\sigma_{jj}}}$, where u_{jk} is the jth entry of $\mathbf{u}_k$.

Proof. (1) $\mathrm{tr}(\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top)=\mathrm{tr}(\boldsymbol{\Lambda}\mathbf{U}^\top\mathbf{U})=\mathrm{tr}\boldsymbol{\Lambda}$, and $\mathrm{Var}(Z_k)=\lambda_k$ by Theorem 7.2.
(2) $\mathrm{Cov}(Z_k,X_j)=\mathbf{u}_k^\top\boldsymbol{\Sigma}\mathbf{e}_j=\lambda_k\mathbf{u}_k^\top\mathbf{e}_j=\lambda_k u_{jk}$, so $\mathrm{Corr}=\lambda_k u_{jk}/(\sqrt{\lambda_k}\sqrt{\sigma_{jj}})$. ■

Theorem 7.4: equicorrelation structure

If $\boldsymbol{\Sigma}=\sigma^2\{(1-\rho)\mathbf{I}_p+\rho\mathbf{1}_p\mathbf{1}_p^\top\}$ with $0<\rho<1$, then $\lambda_1=\sigma^2\{1+(p-1)\rho\}$ with $\mathbf{u}_1=\mathbf{1}_p/\sqrt{p}$, and $\lambda_2=\cdots=\lambda_p=\sigma^2(1-\rho)$ with any orthonormal basis of $\mathbf{1}_p^\perp$ as $\mathbf{u}_2,\dots,\mathbf{u}_p$. The first PC $Z_1=\frac{1}{\sqrt{p}}\sum_j X_j$ is a scaled total and explains the proportion $\rho+(1-\rho)/p$ of the total variance.

Proof. $\boldsymbol{\Sigma}\mathbf{1}_p=\sigma^2\{(1-\rho)+p\rho\}\mathbf{1}_p$, and for $\mathbf{v}\perp\mathbf{1}_p$, $\boldsymbol{\Sigma}\mathbf{v}=\sigma^2(1-\rho)\mathbf{v}$ because $\mathbf{1}_p^\top\mathbf{v}=0$. Since $\mathbb{R}^p=\mathrm{span}(\mathbf{1}_p)\oplus\mathbf{1}_p^\perp$, these are all the eigenpairs, and $1+(p-1)\rho>1-\rho$ for $\rho>0$. Finally $\lambda_1/\mathrm{tr}\boldsymbol{\Sigma}=\{1+(p-1)\rho\}/p=\rho+(1-\rho)/p$. ■

With ρ close to 1, a single component summarizes all p variables; with $\rho=0$ every direction is a principal component and PCA reduces nothing.

Theorem 7.5: Ky Fan maximum principle — q-dimensional subspaces

[Supplement]

$$\max_{\mathbf{V}\in\mathbb{R}^{p\times q},\ \mathbf{V}^\top\mathbf{V}=\mathbf{I}_q}\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})=\lambda_1+\cdots+\lambda_q,$$

attained at $\mathbf{V}=[\mathbf{u}_1,\dots,\mathbf{u}_q]$ (and its rotations).

Proof [Supplement]. $\mathbf{P}=\mathbf{V}\mathbf{V}^\top$ is a rank-q projection with eigenvalues 1 (q times) and 0. By von Neumann's upper bound, $\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})=\mathrm{tr}(\boldsymbol{\Sigma}\mathbf{P})\le\sum_i\lambda_i(\boldsymbol{\Sigma})\lambda_i(\mathbf{P})=\lambda_1+\cdots+\lambda_q$. ■

Remark 7.6: PCA as optimal approximation

[Supplement] Since $\mathbf{I}-\mathbf{V}\mathbf{V}^\top$ is symmetric idempotent, $\mathrm{E}\|(\mathbf{X}-\boldsymbol{\mu})-\mathbf{V}\mathbf{V}^\top(\mathbf{X}-\boldsymbol{\mu})\|^2=\mathrm{tr}\{(\mathbf{I}-\mathbf{V}\mathbf{V}^\top)\boldsymbol{\Sigma}\}=\mathrm{tr}\boldsymbol{\Sigma}-\mathrm{tr}(\mathbf{V}^\top\boldsymbol{\Sigma}\mathbf{V})$, so among q-dimensional subspaces through the mean, the first q PCs minimize the expected squared reconstruction error, the minimum being $\lambda_{q+1}+\cdots+\lambda_p$.

2 Sample Principal Components and the SVD

2.1 Sample PCs

Let $\tilde{\mathbf{X}} = \mathbf{H}_n\mathbf{X} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$ be the centered data matrix, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$, and

$$\hat{\mathbf{u}}_k = \arg\max_{\|\mathbf{u}\|=1,\ \mathbf{u}^\top\hat{\mathbf{u}}_j=0,\ j=1,\dots,k-1} \mathbf{u}^\top\mathbf{S}\mathbf{u}, \qquad \text{scores } t_{ik} = \hat{\mathbf{u}}_k^\top(\mathbf{x}_i - \bar{\mathbf{x}}).$$

The sample PC scores have sample variance $\hat{\lambda}_k$ and are mutually uncorrelated.

Proof [Supplement]. The score vectors are the columns of $\tilde{\mathbf{X}}\hat{\mathbf{U}}$, which are centered because $\mathbf{1}_n^\top\tilde{\mathbf{X}} = \mathbf{0}^\top$. Their sample covariance matrix is $\frac{1}{n-1}\hat{\mathbf{U}}^\top\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}\hat{\mathbf{U}} = \hat{\mathbf{U}}^\top\mathbf{S}\hat{\mathbf{U}} = \hat{\mathbf{\Lambda}}$, which is diagonal with entries $\hat{\lambda}_k$. ■

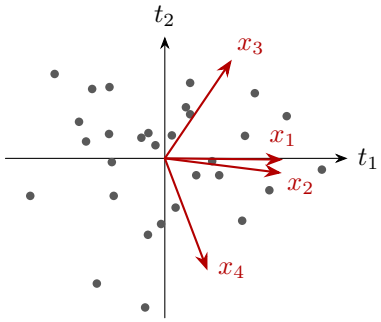


Figure. *Biplot of synthetic standardized data ($n = 30$, $p = 4$; $\hat{\lambda}_1 = 2.29$, $\hat{\lambda}_2 = 1.52$, together 95% of $\mathrm{tr}\mathbf{R} = 4$). Dots are the scores (t_{i1}, t_{i2}); arrow j has coordinates $(\mathrm{Corr}(Z_1, X_j), \mathrm{Corr}(Z_2, X_j))$, scaled by 2.6. x_1, x_2 load only on the first PC; x_3 and x_4 point to opposite sides of the second, which is a contrast between them.*

2.2 Relation to the singular value decomposition

Write the SVD of the $p \times n$ matrix $\tilde{\mathbf{X}}^\top$ $(n > p)$ as

$$\tilde{\mathbf{X}}^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \underbrace{[\mathbf{u}_1 \cdots \mathbf{u}_p]}_{p\times p} \underbrace{\begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_p \end{pmatrix}}_{p\times p} \underbrace{\begin{pmatrix} \mathbf{v}_1^\top \\ \vdots \\ \mathbf{v}_p^\top \end{pmatrix}}_{p\times n}, \qquad \mathbf{U}^\top\mathbf{U} = \mathbf{I}_p,\ \mathbf{V}^\top\mathbf{V} = \mathbf{I}_p,\ d_1 \geq \cdots \geq d_p \geq 0.$$

Then

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{U}\mathrm{diag}\Big(\frac{d_1^2}{n-1}, \dots, \frac{d_p^2}{n-1}\Big)\mathbf{U}^\top, \qquad \begin{pmatrix} \mathbf{z}_{(1)}^\top \\ \vdots \\ \mathbf{z}_{(p)}^\top \end{pmatrix} = \mathbf{Z} = \mathbf{U}^\top\tilde{\mathbf{X}}^\top = \mathbf{D}\mathbf{V}^\top = \begin{pmatrix} d_1\mathbf{v}_1^\top \\ \vdots \\ d_p\mathbf{v}_p^\top \end{pmatrix}.$$

The PC directions are the left singular vectors, the eigenvalues are $d_k^2/(n-1)$, and the kth PC score vector is $\mathbf{z}_{(k)} = \hat{\mathbf{u}}_k^\top\tilde{\mathbf{X}}^\top = d_k\mathbf{v}_k^\top$. Numerically the SVD is preferred to forming $\mathbf{S}$ (computation of PCA).

3 PCA and the Best Approximating Affine Subspace

Motivation. A second view of PCA: the first r components span the r-dimensional affine subspace closest to the data in squared distance. Maximizing variance and minimizing reconstruction error are the same problem.

Theorem 7.7: claim

PCA with r components is equivalent to finding the best r-dimensional affine subspace (hyperplane) approximating the data.

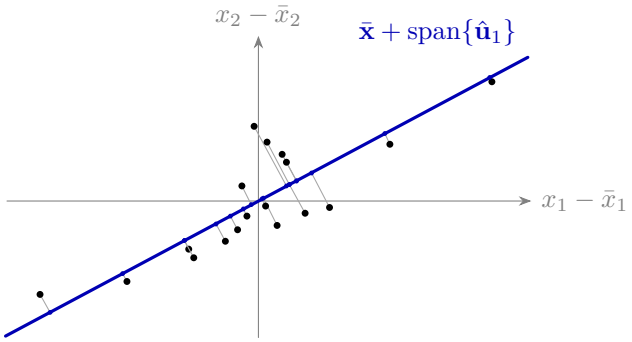


Figure. *Synthetic centered data ($n = 18$, $p = 2$, $r = 1$). Each point $\mathbf{x}_i$ (black) is replaced by its orthogonal projection $\bar{\mathbf{x}} + (\hat{\mathbf{u}}_1^\top \tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_1$ (blue) on the first-PC line, $\hat{\mathbf{u}}_1 = (0.88, 0.47)^\top$. The sum of the squared gray segments is $\text{SSE} = (n-1)\hat{\lambda}_2$, the smallest over all lines; the spread of the blue points along the line is the retained variance $\hat{\lambda}_1$.*

3.1 The problem

Let the data be $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^p$ and $r \in \{1, \ldots, p\}$. We wish to approximate $\mathbf{x}_1, \ldots, \mathbf{x}_n$ by $\bar{\mathbf{x}} + \mathbf{a}_1, \ldots, \bar{\mathbf{x}} + \mathbf{a}_n$ where $\text{rank}([\mathbf{a}_1, \ldots, \mathbf{a}_n]) = r$. The approximation of $\mathbf{x}_i$ by $\bar{\mathbf{x}} + \mathbf{a}_i$ is better when the error $\|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller, and the approximation of $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ by $\{\bar{\mathbf{x}} + \mathbf{a}_i\}$ is better when $\text{SSE} = \sum_{i=1}^n \|\mathbf{x}_i - (\bar{\mathbf{x}} + \mathbf{a}_i)\|_2^2$ is smaller. That is, we wish to find $\mathbf{A}_r = [\mathbf{a}_1, \ldots, \mathbf{a}_n]$ $(p \times n)$ solving

$$(\star) \qquad \min_{\mathbf{A}_r} \sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 \quad \text{subject to} \quad \text{rank}(\mathbf{A}_r) = r.$$

Since every column of a rank-r matrix lies in an r-dimensional subspace, the points $\bar{\mathbf{x}} + \mathbf{a}_i$ lie on an r-dimensional affine subspace through $\bar{\mathbf{x}}$; conversely any such subspace gives feasible $\mathbf{A}_r$.

3.2 Observations

1. With $\tilde{\mathbf{X}}^\top = [\mathbf{x}_1 - \bar{\mathbf{x}}, \ldots, \mathbf{x}_n - \bar{\mathbf{x}}]$ $(p \times n)$,
$$\sum_{i=1}^n \|(\mathbf{x}_i - \bar{\mathbf{x}}) - \mathbf{a}_i\|_2^2 = \|\tilde{\mathbf{X}}^\top - \mathbf{A}_r\|_F^2,$$
because for a $p \times n$ matrix $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_n]$ the Frobenius norm is $\|\mathbf{B}\|_F^2 = \sum_{i=1}^p \sum_{j=1}^n b_{ij}^2 = \sum_{j=1}^n \|\mathbf{b}_j\|_2^2 = \text{tr}(\mathbf{B}\mathbf{B}^\top) = \text{tr}(\mathbf{B}^\top\mathbf{B})$.
2. The sample PCA of $(\mathbf{x}_1, \ldots, \mathbf{x}_n)$ is given by
   (1) the eigendecomposition $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top = \sum_{i=1}^p \hat{\lambda}_i \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{\lambda}_i$ the ith PC variance, $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_r > \hat{\lambda}_{r+1} \geq \cdots \geq \hat{\lambda}_p$;
   (2) the SVD $\tilde{\mathbf{X}}^\top_{(p\times n)} = \hat{\mathbf{U}}\hat{\mathbf{D}}\hat{\mathbf{V}}^\top = \sum_{i=1}^p \hat{d}_i \hat{\mathbf{u}}_i \hat{\mathbf{v}}_i^\top$, where $\hat{\mathbf{u}}_i$ is the ith PC direction vector and $\hat{d}_i\hat{\mathbf{v}}_i$ the ith PC score vector;
   (3) $\hat{\mathbf{u}}_i^\top \tilde{\mathbf{X}}^\top = \hat{d}_i \hat{\mathbf{v}}_i^\top$, i.e.
$$\begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix}.$$

Theorem 7.8: solution (Eckart–Young–Mirsky)

The $\mathbf{A}_r$ that minimizes $(\star)$ is

$$\mathbf{A}_r = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r][\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]^\top \tilde{\mathbf{X}}^\top,$$

the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace, and the minimum is $\sum_{i>r} \hat{d}_i^2 = (n-1)\sum_{i=r+1}^{p} \hat{\lambda}_i$.

Proof. Step 1 (parametrize rank-r matrices). Since $\operatorname{rank}(\mathbf{A}_r) = r$, we can write $\mathbf{A}_r = \mathbf{E}\mathbf{C}^\top$ with $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_r]$ $(p \times r)$ satisfying $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_r$ (an orthonormal basis of the column space of $\mathbf{A}_r$) and $\mathbf{C} = \begin{pmatrix} \mathbf{c}_1^\top \\ \vdots \\ \mathbf{c}_n^\top \end{pmatrix}$ an $n \times r$ matrix. Writing $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, $(\star)$ is equivalent to $\min_{\mathbf{E},\mathbf{C}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2$, where

$$(\star\star) \qquad \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{C}^\top\|_F^2 = \big\|(\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_n) - \mathbf{E}[\mathbf{c}_1, \dots, \mathbf{c}_n]\big\|_F^2 = \sum_{i=1}^{n} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2.$$

Step 2 (optimal $\mathbf{C}$ for fixed $\mathbf{E}$). For any given $\mathbf{E}$, $(\star\star)$ separates over i and its minimizer is

$$\hat{\mathbf{c}}_i = \arg\min_{\mathbf{c}_i} \|\tilde{\mathbf{x}}_i - \mathbf{E}\mathbf{c}_i\|_2^2 = (\mathbf{E}^\top\mathbf{E})^{-1}\mathbf{E}^\top\tilde{\mathbf{x}}_i = \mathbf{E}^\top\tilde{\mathbf{x}}_i,$$

the projection scores of $\tilde{\mathbf{x}}_i$ onto $\operatorname{span}(\mathbf{E})$ (least squares). Stacking, $\hat{\mathbf{C}} = \begin{pmatrix} \tilde{\mathbf{x}}_1^\top\mathbf{E} \\ \vdots \\ \tilde{\mathbf{x}}_n^\top\mathbf{E} \end{pmatrix} = \tilde{\mathbf{X}}\mathbf{E}$, i.e. $\hat{\mathbf{C}}^\top = \mathbf{E}^\top\tilde{\mathbf{X}}^\top$.

Step 3 (profile out $\mathbf{C}$). Plugging $\hat{\mathbf{C}}$ into $(\star\star)$ (profiling, Theorem 2.26), and using that $\mathbf{I} - \mathbf{E}\mathbf{E}^\top$ is symmetric idempotent, $(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)^2 = \mathbf{I} - \mathbf{E}\mathbf{E}^\top$,

$$\begin{aligned}
\min_{\mathbf{E}} \|\tilde{\mathbf{X}}^\top - \mathbf{E}\mathbf{E}^\top\tilde{\mathbf{X}}^\top\|_F^2 &= \min_{\mathbf{E}} \|(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top\|_F^2 = \min_{\mathbf{E}} \operatorname{tr}(\tilde{\mathbf{X}}(\mathbf{I} - \mathbf{E}\mathbf{E}^\top)\tilde{\mathbf{X}}^\top) \\
&= \min_{\mathbf{E}} \Big\{ \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\operatorname{tr}\Big(\mathbf{E}^\top \frac{\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}}{n-1}\mathbf{E}\Big) \Big\} \\
&= \operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) - (n-1)\max_{\mathbf{E}} \operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top).
\end{aligned}$$

Step 4 (von Neumann). Recall $\mathbf{S} = \hat{\mathbf{U}}\hat{\mathbf{\Lambda}}\hat{\mathbf{U}}^\top$. $\mathbf{E}\mathbf{E}^\top$ is symmetric with eigenvalues 1 (r times) and 0 ($p - r$ times), so von Neumann's trace inequality (Theorem 2.14) gives

$$\operatorname{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top) \le \hat{\lambda}_1 + \cdots + \hat{\lambda}_r,$$

with equality when $\mathbf{E} = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r]$ (or any $\mathbf{E}$ spanning the same space).

Step 5 (the minimizer). Therefore

$$\mathbf{A}_r = \mathbf{E}\hat{\mathbf{C}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{\mathbf{u}}_1^\top \\ \vdots \\ \hat{\mathbf{u}}_r^\top \end{pmatrix} \tilde{\mathbf{X}}^\top = [\hat{\mathbf{u}}_1, \dots, \hat{\mathbf{u}}_r] \begin{pmatrix} \hat{d}_1 & & \\ & \ddots & \\ & & \hat{d}_r \end{pmatrix} \begin{pmatrix} \hat{\mathbf{v}}_1^\top \\ \vdots \\ \hat{\mathbf{v}}_r^\top \end{pmatrix} = \sum_{i=1}^{r} \hat{\mathbf{u}}_i \hat{d}_i \hat{\mathbf{v}}_i^\top$$

by Observation 2(3): the projection of $\tilde{\mathbf{X}}^\top$ onto the first r PC subspace. Since $\operatorname{tr}(\tilde{\mathbf{X}}\tilde{\mathbf{X}}^\top) = (n-1)\operatorname{tr}\mathbf{S} = (n-1)\sum_i \hat{\lambda}_i$, the minimum is $(n-1)\sum_{i>r} \hat{\lambda}_i = \sum_{i>r} \hat{d}_i^2$. ∎

Remark 7.9: interpretation

[Supplement]

- The best approximating hyperplane is $\bar{\mathbf{x}}+\text{span}\{\hat{\mathbf{u}}_1,\dots,\hat{\mathbf{u}}_r\}$ and each $\mathbf{x}_i$ is approximated by its orthogonal projection $\bar{\mathbf{x}}+\sum_{k\le r}(\hat{\mathbf{u}}_k^\top\tilde{\mathbf{x}}_i)\hat{\mathbf{u}}_k$: the PC scores are the coordinates on the hyperplane.
- The fitted values satisfy $\hat{\mathbf{c}}_i=\mathbf{E}^\top\tilde{\mathbf{x}}_i$, i.e. reconstruction uses projection scores, exactly the decomposition $\mathbf{x}=\bar{\mathbf{x}}+\sum s_{\mathbf{v}_j}(\mathbf{x}-\bar{\mathbf{x}})\mathbf{v}_j+r_{\mathcal{L}_q}(\mathbf{x})$ of Ch. 1.
- Minimizing squared error and maximizing retained variance (the $\arg\max\mathbf{u}^\top\mathbf{S}\mathbf{u}$ definition) are the same problem, since $\text{SSE}=(n-1)\{\text{tr}\mathbf{S}-\text{tr}(\mathbf{S}\mathbf{E}\mathbf{E}^\top)\}$.
- If $\hat{\lambda}_r=\hat{\lambda}_{r+1}$ the optimal subspace is not unique; the strict gap $\hat{\lambda}_r>\hat{\lambda}_{r+1}$ makes it unique.

3.3 Correlation PCA

If the units differ, use PCA of standardized data: $\mathbf{R}=\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}\boldsymbol{\Sigma}\mathbf{D}_{\boldsymbol{\Sigma}}^{-1/2}=\mathbf{U}_R\boldsymbol{\Lambda}_R\mathbf{U}_R^\top$ (sample: $\mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$). Since $\text{tr}\mathbf{R}=p$, the proportion explained is λ_k^R/p. Covariance and correlation PCs are in general not simple transformations of each other (scale dependence).

4 How Many Components to Keep?

- **Scree plot / cumulative scree plot**: plot $\hat{\lambda}_k$ against k and look for the elbow, or take the q at which the cumulative proportion reaches a target (e.g. 80–90%).
- **Kaiser's rule**: keep components with $\hat{\lambda}_k>\bar{\lambda}=\frac{1}{p}\sum_j\hat{\lambda}_j$; for correlation PCA, $\hat{\lambda}_k>1$.
- **Likelihood ratio testing**: under normality test $H_0(k):\lambda_{k+1}=\cdots=\lambda_p$ (the remaining space is spherical). With arithmetic mean $\bar{a}$ and geometric mean $\bar{g}$ of the last $p-k$ sample eigenvalues,

$$-2\log\Lambda=n(p-k)\log\frac{\bar{a}}{\bar{g}}\ \dot{\sim}\ \chi^2_{(p-k+2)(p-k-1)/2};$$

since $\bar{a}\ge\bar{g}$ (AM–GM), the statistic is small when the eigenvalues are nearly equal.

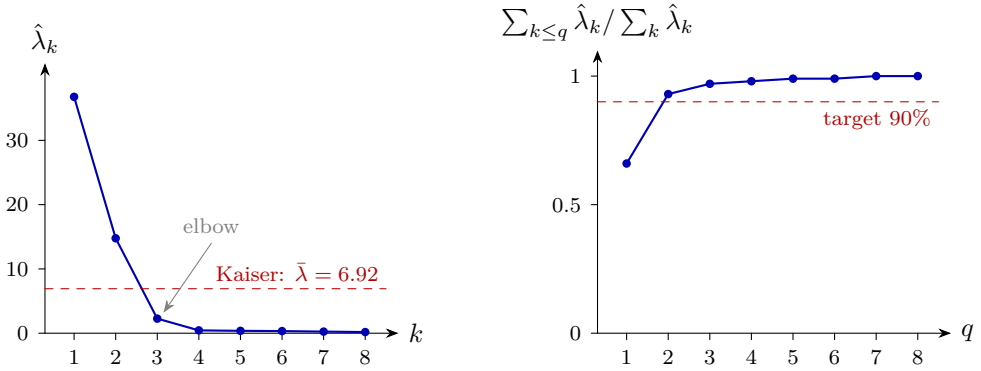


Figure. *Scree plot (left) and cumulative scree plot (right) for synthetic data ($n=200$, $p=8$, generated with three strong directions). The elbow is at $k=3$, Kaiser's rule keeps $q=2$ ($\hat{\lambda}_1,\hat{\lambda}_2>\bar{\lambda}$), and the 90% target is reached at $q=2$ (93%).*

Remark 7.10: cautions in interpretation

- The sign of an eigenvector is arbitrary; what matters is the sign pattern (same signs: "size"; mixed signs: "contrast"). In the scores example $\mathbf{u}_1 \propto (1,1)$ was overall level and $\mathbf{u}_2 \propto (1,-1)$ the contrast between subjects.
- With tied eigenvalues the directions within the eigenspace are not unique.
- Sample eigenvalues are biased: large ones are overestimated and small ones underestimated. Under normality, for distinct λ_k, $\sqrt{n}(\hat{\lambda}_k - \lambda_k) \Rightarrow N(0, 2\lambda_k^2)$.
- High explained variance does not mean usefulness for classification or prediction (the $X_1 = 10Z$, $X_2 = Y$ example of Ch. 1).

5 Example: Scores in Two Subjects Revisited

For $\mathbf{S} = \begin{pmatrix} 125 & 75 \\ 75 & 125 \end{pmatrix}$ the characteristic equation $(125-\lambda)^2 - 75^2 = 0$ gives $\lambda = 200, 50$; the null space of $\mathbf{S} - 200\mathbf{I} = \begin{pmatrix} -75 & 75 \\ 75 & -75 \end{pmatrix}$ is spanned by $(1,1)^\top$. $\mathrm{Corr}(Z_1, X_1) = u_{11}\sqrt{\lambda_1}/\sqrt{s_{11}} = \frac{1}{\sqrt{2}}\sqrt{200}/\sqrt{125} = 0.894$, $\mathrm{Corr}(Z_2, X_1) = \frac{1}{\sqrt{2}}\sqrt{50}/\sqrt{125} = 0.447$, and $0.894^2 + 0.447^2 = 1$: each variable's variance is fully decomposed over the PCs.

6 Selected exercises

Motivation. PCA is worthwhile only if the variables are correlated, and the eigenvalue tests above compare a covariance matrix with a spherical one. The first two exercises derive the corresponding likelihood ratio tests; the third is a covariance whose components can be found by hand.

Exercise 7.1 (Testing that all variables are independent; J&W 8.9(a))

Let $\mathbf{x}_1, \ldots, \mathbf{x}_n$ be a sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $n > p$. Show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \mathrm{diag}(\sigma_{11}, \ldots, \sigma_{pp})$ is

$$\Lambda = \frac{|\mathbf{S}|^{n/2}}{\prod_{j=1}^{p} s_{jj}^{n/2}} = |\mathbf{R}|^{n/2},$$

that $0 < \Lambda \le 1$, and that $-2\log\Lambda$ is approximately $\chi^2_{p(p-1)/2}$ for large n.

Solution.

Step 1 (unrestricted maximum). By Theorem 4.7 and the remark after it, $\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L = (2\pi)^{-np/2}|\mathbf{S}_0|^{-n/2}e^{-np/2}$ with $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$.

Step 2 (maximum under H_0). If $\boldsymbol{\Sigma}$ is diagonal, the density of $\mathbf{x}_i$ is the product of univariate $N(\mu_j, \sigma_{jj})$ densities, so $L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \prod_{j=1}^{p} L_j(\mu_j, \sigma_{jj})$ with L_j the likelihood of the jth coordinate. The factors have separate parameters, so each is maximized separately; Theorem 4.7 with $p = 1$ gives $\hat{\mu}_j = \bar{x}_j$, $\hat{\sigma}_{jj} = (\mathbf{S}_0)_{jj}$ and

$$\max_{H_0} L = \prod_{j=1}^{p} (2\pi)^{-n/2}(\mathbf{S}_0)_{jj}^{-n/2}e^{-n/2} = (2\pi)^{-np/2}\Big\{\prod_{j}(\mathbf{S}_0)_{jj}\Big\}^{-n/2}e^{-np/2}.$$

Step 3 (the ratio). Dividing,

$$\Lambda = \Big(\frac{|\mathbf{S}_0|}{\prod_j (\mathbf{S}_0)_{jj}}\Big)^{n/2} = \Big(\frac{|\mathbf{S}|}{\prod_j s_{jj}}\Big)^{n/2},$$

since $|\mathbf{S}_0| = (\frac{n-1}{n})^p |\mathbf{S}|$ and $\prod_j (\mathbf{S}_0)_{jj} = (\frac{n-1}{n})^p \prod_j s_{jj}$. With $\mathbf{D} = \operatorname{diag}(s_{11}, \dots, s_{pp})$, $\mathbf{R} = \mathbf{D}^{-1/2}\mathbf{S}\mathbf{D}^{-1/2}$ and $|\mathbf{R}| = |\mathbf{S}| / \prod_j s_{jj}$.

Step 4. $\mathbf{S} \succ 0$ a.s. (as $n > p$), so $\mathbf{R} \succ 0$ and Exercise 2.2 gives $0 < |\mathbf{R}| \le 1$, with $\Lambda = 1$ iff $\mathbf{R} = \mathbf{I}_p$.

Step 5 (degrees of freedom). The unrestricted model has $p + p(p+1)/2$ free parameters and H_0 has $2p$; the large-sample likelihood ratio theory used in Ch. 5 gives $-2\log\Lambda \overset{\cdot}{\sim} \chi^2_\nu$ with $\nu = p(p+1)/2 - p = p(p-1)/2$, the number of off-diagonal correlations. ■

Exercise 7.2 (Sphericity test; J&W 8.9(b))

In the setting of Exercise 7.1, show that the likelihood ratio statistic for $H_0 : \boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_p$ ($\sigma^2 > 0$ unknown) is

$$\Lambda = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2} = \Big(\frac{\bar{g}}{\bar{a}}\Big)^{np/2},$$

where $\bar{a}$ and $\bar{g}$ are the arithmetic and geometric means of the eigenvalues $\hat{\lambda}_1, \dots, \hat{\lambda}_p$ of $\mathbf{S}$. Show $-2\log\Lambda \ge 0$ and find its degrees of freedom; compare with the test of $H_0(k)$ in the last section.

Solution.

Step 1 (log-likelihood under H_0). With $Q(\boldsymbol{\mu}) = \sum_i \|\mathbf{x}_i - \boldsymbol{\mu}\|^2$,

$$\log L(\boldsymbol{\mu}, \sigma^2 \mathbf{I}) = -\frac{np}{2}\log(2\pi\sigma^2) - \frac{Q(\boldsymbol{\mu})}{2\sigma^2}.$$

$Q(\boldsymbol{\mu}) = Q(\bar{\mathbf{x}}) + n\|\bar{\mathbf{x}} - \boldsymbol{\mu}\|^2$ is minimized at $\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}}$ for every σ^2, and $Q(\bar{\mathbf{x}}) = n\operatorname{tr}\mathbf{S}_0$.

Step 2 ($\hat{\sigma}^2$). Let $c = Q(\bar{\mathbf{x}})/(np) = \operatorname{tr}\mathbf{S}_0/p$ and $x = c/\sigma^2$. Then

$$\log\sigma^2 + \frac{Q(\bar{\mathbf{x}})}{np\,\sigma^2} = \log c - \log x + x \;\ge\; \log c + 1$$

by Theorem 2.15, with equality iff $x = 1$. Hence $\hat{\sigma}^2 = \operatorname{tr}\mathbf{S}_0/p$ and $\max_{H_0} L = (2\pi\hat{\sigma}^2)^{-np/2} e^{-np/2}$.

Step 3 (the ratio). With Step 1 of Exercise 7.1,

$$\Lambda = \frac{|\mathbf{S}_0|^{n/2}}{(\operatorname{tr}\mathbf{S}_0/p)^{np/2}} = \Big(\frac{|\mathbf{S}|}{(\operatorname{tr}\mathbf{S}/p)^p}\Big)^{n/2},$$

because numerator and denominator are both homogeneous of degree $np/2$ in $\mathbf{S}_0$. Since $|\mathbf{S}| = \prod_i \hat{\lambda}_i = \bar{g}^p$ and $\operatorname{tr}\mathbf{S}/p = \bar{a}$, $\Lambda = (\bar{g}/\bar{a})^{np/2}$.

Step 4. $-2\log\Lambda = np\log(\bar{a}/\bar{g}) \ge 0$ by the AM–GM inequality, with equality iff all $\hat{\lambda}_i$ are equal.

Step 5 (degrees of freedom). The free parameters are $p + p(p+1)/2$ versus $p+1$ under H_0, so $\nu = p(p+1)/2 - 1 = (p+2)(p-1)/2$. This is the statistic $n(p-k)\log(\bar{a}/\bar{g})$ with $\chi^2_{(p-k+2)(p-k-1)/2}$ of the last section for $k = 0$; for $k \ge 1$ the same comparison is applied to the last $p-k$ eigenvalues. ■

Exercise 7.3 (A tridiagonal covariance; J&W 8.4)

Find the principal components and the proportion of total variance explained by each for

$$\boldsymbol{\Sigma} = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix}, \qquad 0 < \rho < \frac{1}{\sqrt{2}}.$$

Solution.

Step 1 (eigenvalues). Let $\mathbf{C} = \boldsymbol{\Sigma}/\sigma^2$. Expanding along the first row,

$$|\mathbf{C} - \lambda \mathbf{I}| = (1-\lambda)\{(1-\lambda)^2 - \rho^2\} - \rho \cdot \rho(1-\lambda) = (1-\lambda)\{(1-\lambda)^2 - 2\rho^2\},$$

so the eigenvalues of $\boldsymbol{\Sigma}$ are $\sigma^2(1+\sqrt{2}\rho) > \sigma^2 > \sigma^2(1-\sqrt{2}\rho)$, all positive because $\rho < 1/\sqrt{2}$.

Step 2 (eigenvectors). For $\lambda = 1 \pm \sqrt{2}\rho$ the first and third rows of $(\mathbf{C} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$ read $\mp\sqrt{2}\rho x_1 + \rho x_2 = 0$ and $\rho x_2 \mp \sqrt{2}\rho x_3 = 0$, so $x_2 = \pm\sqrt{2}x_1$ and $x_3 = x_1$. For $\lambda = 1$ they give $x_2 = 0$ and the middle row gives $x_1 = -x_3$. Normalizing,

$$\mathbf{u}_1 = \tfrac{1}{2}(1, \sqrt{2}, 1)^\top, \qquad \mathbf{u}_2 = \tfrac{1}{\sqrt{2}}(1, 0, -1)^\top, \qquad \mathbf{u}_3 = \tfrac{1}{2}(1, -\sqrt{2}, 1)^\top.$$

Step 3 (components). $Z_1 = \frac{1}{2}(X_1 + \sqrt{2}X_2 + X_3)$, $Z_2 = \frac{1}{\sqrt{2}}(X_1 - X_3)$, $Z_3 = \frac{1}{2}(X_1 - \sqrt{2}X_2 + X_3)$. Since $\mathrm{tr}\boldsymbol{\Sigma} = 3\sigma^2$, the proportions are $(1+\sqrt{2}\rho)/3$, $1/3$ and $(1-\sqrt{2}\rho)/3$.

Z_1 is a weighted level with the middle variable counted most; Z_2 contrasts the two ends and does not depend on ρ. As $\rho \uparrow 1/\sqrt{2}$, $\mathrm{Var}(Z_3) \to 0$ and $\boldsymbol{\Sigma}$ becomes singular. ■